Example 5.12

Draw the graph of $f(x) = a^x$, $a \neq 1$ and $a > 0$

Solution

We know that, domain set is R, range set is $(0, \infty)$ and the curve passing through the point (0, 1)

Case (i) when $a > 1$

$$y = f(x) = a^x = \begin{cases} < 1 & \text{if } x < 0 \\ = 1 & \text{if } x = 0 \\ > 1 & \text{if } x > 0 \end{cases}$$

We noticed that as x increases, y is also increases and $y > 0$.

$\therefore$ The graph is as shown in the figure.

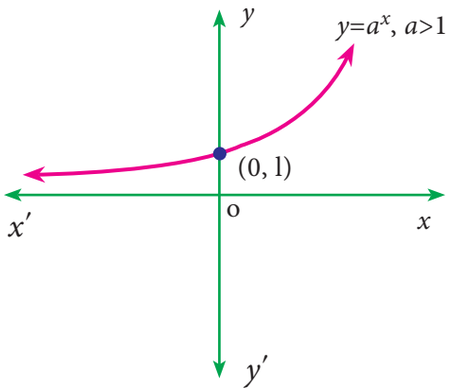


Fig 5.6

Case (ii) when $0 < a < 1$

$$y = f(x) = a^x = \begin{cases} > 1 & \text{if } x < 0 \\ = 1 & \text{if } x = 0 \\ < 1 & \text{if } x > 0 \end{cases}$$

We noticed that as x increases, y decreases and $y > 0$.

$\therefore$ The graph is as shown in the figure.

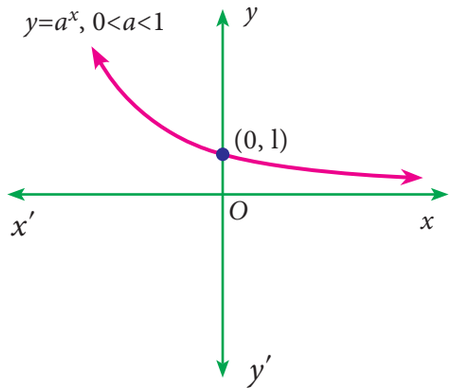


Fig 5.7

NOTE

(i) *The value of e lies between 2 and 3. i.e. $2 < e < 3$*

(ii) *The graph of $f(x) = e^x$ is identical to that of $f(x) = a^x$ if $a > 1$ and the graph of $f(x) = e^{-x}$ is identical to that of $f(x) = a^x$ if $0 < a < 1$*

Example 5.13

Draw the graph of $f(x) = \log_a x$; $x > 0$, $a > 0$ and $a \neq 1$.

Solution

We know that, domain set is $(0, \infty)$, range set is R and the curve passing the point (1, 0)

Case (i) when $a > 1$ and $x > 0$

$$f(x) = \log_a x = \begin{cases} < 0 & \text{if } 0 < x < 1 \\ = 0 & \text{if } x = 1 \\ > 0 & \text{if } x > 1 \end{cases}$$

We noticed that as x increases, $f(x)$ is also increases.

$\therefore$ The graph is as shown in the figure.

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